Unit 7 Test 2 Review — ANSWER KEY

Part 1: Roots and Factors

1. Roots: $\mathbf{x = 1, 2, 3}$
Work: Factors to $(x-1)(x-2)(x-3)$. *Graph has points at* $x = 1, 2, 3$.

2. A, C
Work: $f(x)$ *factors to* $(x-1)(x-5)(x+3)$. *Roots are* $1, 5, -3$.

3.

- Factors: $\mathbf{(x-1), (x+3), (x-3), (x^2-1)}$
- Not Factors: $\mathbf{(x^2+9), (x+1)}$

4. C (Both are factors)
Work: $x^4 - 625 = (x^2-25)(x^2+25)$.

5. Alex is correct.
Explanation: Sam failed to factor the difference of squares (x^2-16) *completely.*

Part 2: Factoring by Grouping

6. $\mathbf{(x-5)(x+5)(2x+1)}$ or $(x^2-25)(2x+1)$

7. Linear: $\mathbf{2x-3}$ Quadratic: $\mathbf{3x^2+1}$
Note: Original expression $6x^3 - 9x^2 + 2x - 3$ *(assuming sign fix based on grid options) factors to* $(2x-3)(3x^2+1)$. *If strict to paper, check signs.*

8. B $(x+7)$**, D** (x^2+2)

Part 3: Sum & Difference of Cubes

9. B $(3x+2y^2)$ (Note: Typo in question options? Usually $27x^3+8y^6 = (3x+2y^2)(9x^2-6xy^2+4y^4)$.)
Correct Selection from list: **A** $(3x+2y^2)$ and **D** $(9x^2-6xy^2+4y^4)$.

10. $\mathbf{(2x-3y)(4x^2+6xy+9y^2)}$

11. $\mathbf{(x^2+9)(x^4-9x^2+81)}$

Part 4: Rational Root Theorem

12.

- a) Possible Roots: $\mathbf{\pm 1, \pm 3, \pm \frac{1}{2}, \pm \frac{3}{2}}$
- b/c) Factored: $\mathbf{(x+1)(2x-1)(x-3)}$

13.

- a) Possible Roots: $\mathbf{\pm 1, \pm 2, \pm 4, \pm \frac{1}{3}, \pm \frac{2}{3}, \pm \frac{4}{3}}$
- b) Factored: $\mathbf{(x-2)(x+2)(3x-1)}$

Part 5: Mixed Practice

14. Strategy: Grouping Answer: $\mathbf{(x-3)(x+3)(x+5)}$

15. Strategy: Sum of Cubes Answer: $\mathbf{(6x+7)(36x^2-42x+49)}$

16. Strategy: Grouping Answer: $\mathbf{(5x^2-2)(x-3)}$

BONUS: A $\mathbf{(x-1)(x^2+10x+37)}$